

Q1. (a) Analyze the effectiveness of cohesive devices and signpost words in enhancing readability of technical documents

Technical documents contain a lot of information and complex ideas. **Cohesive devices and signpost words** help connect ideas and guide the reader. They make technical documents clear, logical and easy to read.

1. Cohesive Devices - Cohesive devices are words or techniques that **connect sentences and ideas**.

Examples:

- **Pronouns:** it, they, this, these
- **Conjunctions:** and, but, because, so
- **Repeated keywords:** repeating important terms
- **Synonyms:** using words with similar meanings

Effectiveness of Cohesive Devices

1. Smooth flow between sentences.
2. Avoid unnecessary repetition.
3. keep all sentences connected to the main topic.
4. Readers can understand the information easily.

2. Signpost Words - Signpost words show the **relationship between ideas** and guide the reader.

Purpose	Signpost Words
Addition	also, furthermore, moreover
Contrast	however, but, whereas
Cause/Effect	therefore, because, as a result
Example	for example, for instance
Conclusion	finally, in conclusion

Example

The system saves electricity. **Furthermore**, it reduces operating costs. **For example**, the automatic control system switches off unnecessary devices. **Therefore**, the system is suitable for energy-efficient buildings.

Here, “**Furthermore**,” “**For example**,” and “**Therefore**” make the relationship between ideas clear.

Q1. (b) Differentiate technical writing and general writing with appropriate examples

Technical writing communicates technical or specialized information, while **general writing** communicates ideas, feelings or information to a wider audience.

Difference Between Technical Writing and General Writing

Basis	Technical Writing	General Writing
Purpose	To inform, explain or instruct	To entertain, express or inform
Audience	Technical or specialized readers	General public
Language	Formal, precise and simple	Can be informal, creative or expressive
Content	Facts, data, procedures and instructions	Stories, opinions, experiences and ideas
Style	Objective and direct	Can be personal and emotional
Structure	Usually systematic and organized	Structure can be flexible
Examples	Project report, user manual, research paper	Novel, story, blog, personal essay
Main Focus	Accuracy and clarity	Expression and engagement

Example 1 – Technical Writing

“The temperature sensor measures the soil temperature and sends the data to the microcontroller.”

This is technical writing because it gives **specific information about a system**.

Example 2 – General Writing

“The farmer was happy to see his crops growing well after the rain.”

This is general writing because it expresses an **experience and feeling**.

Q2. Importance of Project Reports in STEM Fields

A **project report** is a formal document that explains a project from its **problem identification to the final result**. In STEM fields — **Science, Technology, Engineering and Mathematics** — project reports are important because they record the work, methods, results and conclusions in a systematic way.

1. Documents the Project Work

- A project report provides a **complete record** of the project.
- It includes:
 - Problem statement
 - Objectives
 - Methodology
 - Results
 - Conclusion

2. Explains the Problem and Objectives

- A report clearly explains **what problem is being solved and what the project aims to achieve**.
- **Example:** An AI project may aim to develop a system that **predicts crop diseases using leaf images**.

3. Presents the Methodology

- It explains the **methods, tools, technologies and procedures** used to complete the project.
 - **Example:** A software project report may mention **Python, machine learning algorithms, dataset preparation and model training**.
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4. Records Results and Findings

- Project reports present the results obtained from experiments, testing or analysis.

→ Results can be shown using:

- Tables
- Graphs
- Charts
- Images

Example: An IoT project can show the **amount of water saved by an automatic irrigation system**.

5. Helps in Evaluation

→ Teachers, researchers and project reviewers can use the report to **evaluate the quality and performance of the project**.

6. Supports Scientific and Engineering Communication

STEM professionals need to communicate technical information clearly. A project report develops **technical writing and communication skills**.

Example: An engineer can use a report to explain a new **bridge design** to other engineers.

7. Provides Evidence for the Project

→ A report provides written evidence of the work performed and the results achieved.

→ **Example:** A researcher can use experimental results in a report to support the effectiveness of a **new water purification method**.

8. Helps Identify Problems and Limitations

→ A good report explains the difficulties and limitations faced during the project.

→ This helps future researchers find better solutions.

9. Supports Decision Making

- Project reports provide facts and results that help organizations make better technical decisions.
 - **Example:** A company can study a report comparing two **energy-efficient machines** before purchasing one.
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10. Creates a Permanent Reference

- A project report acts as a **permanent record** that can be referred to later.
- It can help during:
 - Project maintenance
 - Future development
 - Research
 - Auditing
 - Academic evaluation